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Asymmetric Catalysis

Enantio- and Diastereoselective Synthesis of Chiral Tetrasubstituted α-Amino Allenoates Bearing a Vicinal All-Carbon Quaternary Stereocenter with Dual-Copper-Catalysis

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Abstract: The skeletons of chiral tetrasubstituted allenes bearing a vicinal all-carbon quaternary stereocenter are of importance but still challenging to synthesize. Herein, we report enantio- and diastereoselective γ-additions of 1-alkynyl ketimines with dual-copper-catalysis under mild conditions, affording chiral tetrasubstituted α-amino allenoates bearing a vicinal all-carbon quaternary stereocenter in high yields (up to 99% yield) with excellent enantioselectivities (up to 99% *ee*) and diastereoselectivities (up to >20:1 *dr*). Importantly, the stereodivergent synthesis of the products was realized by the asymmetric γ-addition reaction and the Grignard reagent promoted epimerization. Moreover, the dual-copper-catalyzed γ-addition reactions were smoothly applied to a gram-scale reaction and adopted to introduce chiral tetrasubstituted allenyl moieties into bioactive molecules. Mechanistic experiments and density functional theory (DFT) calculations demonstrated that the asymmetric γ-addition reactions were catalyzed by double chiral copper catalysts.

Introduction

Chirality is one of the elemental natures of substances, producing a world of diversity. For example, allene-based axial chirality and carbon-based central chirality are widespread in natural products and pharmaceuticals.[1,2] Compared with molecules containing a single type of chirality, compounds possessing both axial and central chiralities, particularly chiral allenes bearing a vicinal stereocenter including chiral tetrasubstituted allenes with a vicinal all-carbon quaternary center, have not attracted much attention, although they are also found in numerous bioactive molecules and represent one of the most significant categories (Figure 1).[1,3] Over the past decade, a variety of methods for the synthesis of optically axial chiral allenes bearing a vicinal stereocenter have been reported successively.[4–6] However, because the construction of a single chiral tetrasubstituted allene or all-carbon substituted quaternary center remains challenging,[7–9] the asymmetric synthesis of chiral tetrasubstituted allenes with a vicinal all-carbon quaternary center are much more challenging to forge and the asymmetric synthetic methods are still in infancy due to their increasing structural congestion and chiral complexity (Scheme 1a).[10]

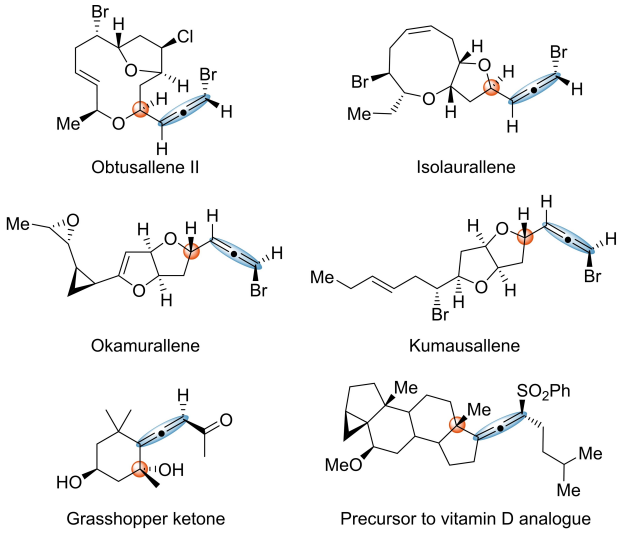


Figure 1. Chiral allenes bearing a vicinal stereocenter in natural products and pharmaceuticals.

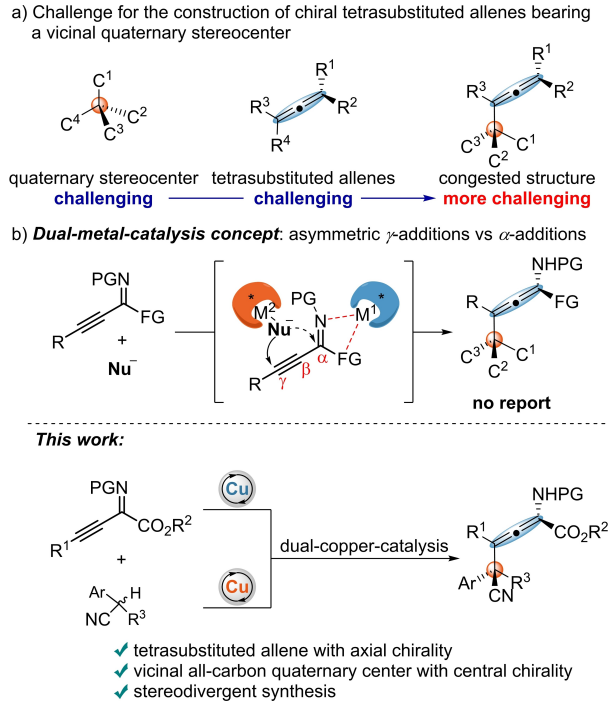
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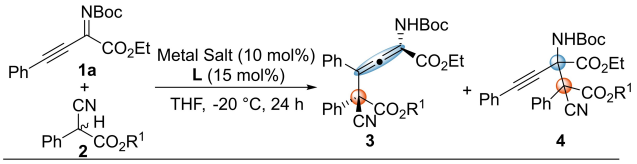
Scheme 1. Construction of chiral tetrasubstituted allenes bearing a vicinal quaternary stereocenter.

1-Alkynyl ketimines are important synthons containing multi active sites (α, β and γ).[11] The γ-selective addition of this type of substrates will afford allenamines, which has attracted great attention for their considerable antiretroviral activities and synthetic applications.[1b,12] Based on previous work concerning dual-metal-catalysis,[13,14] we envisioned a concept of asymmetric γ-additions of 1-alkynyl ketimines to synthesize tetrasubstituted allenes with a vicinal all-carbon quaternary stereocenter using dual-metal-catalysis, in which M^2-ligand can be used to activate nucleophiles and form repulsion with M^1-ligand-activated imine unit to prevent the α-additions (Scheme 1b, top). In the realized catalytic protocol, readily available α-substituted cyanoacetates are applying as the nucleophiles for the successful generation of both the fully loaded axial allene and vicinal acyclic all-carbon quaternary stereocenter using dual-metal-catalysis with the same copper catalyst (Scheme 1b, bottom).

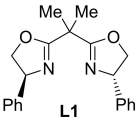
Results and Discussion

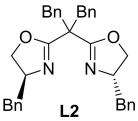
Based on the above hypothesis of dual-metal-catalysis, our study commenced with the investigation of several combinations of transition metal catalysts. Considering the efficacious catalytic outcome of copper salts in the asymmetric addition reactions of 1-alkynyl ketimines,[11d] we selected $Cu(OAc)_2$ as the catalyst to control ketimine substrate **1a** and several other transition metal salts to chelate α-substituted cyanoacetate **2** (Table 1, entries 1–3). To our delight, by using bis(oxazoline) (BOX) **L1** as the ligand, all the reactions afforded high γ-selectivities. Subsequently, the

Table 1: Optimization of reaction conditions.[a]



entry	R^1	Metal Salt	L	yield (%)[b] (3/4)	dr[c]	ee (%)[d]
1[e]	Me	$Cu(OAc)_2$+$Ni(OAc)_2$	L1	80/trace	1.2:1	28/-4
2[e]	Me	$Cu(OAc)_2$+$Zn(OAc)_2$	L1	70/6	1.3:1	47/34
3[e]	Me	$Cu(OAc)_2$+$Co(OAc)_2$	L1	57/trace	1.6:1	55/36
4	Me	$Cu(OAc)_2$	L1	98/N.D.	1.6:1	23/35
5	Me	$Ni(OAc)_2$	L1	67/13	1.2:1	21/-14
6	Me	$Zn(OAc)_2$	L1	29/30	1.6:1	40/6
7	Me	$Co(OAc)_2$	L1	50/22	1.7:1	23/-28
8	Et	$Cu(OAc)_2$	L1	92/N.D.	1.1:1	30/35
9	Bn	$Cu(OAc)_2$	L1	98/N.D.	1.5:1	34/26
10	^{t}Bu	$Cu(OAc)_2$	L1	91/N.D.	2.8:1	52/49
11[f,g]	^{t}Bu	CuDPP[h]	L1	91/N.D.	4.8:1	83/69
12[f,g]	^{t}Bu	CuDPP	L2	99/N.D.	>20:1	95
13[f,i]	^{t}Bu	CuDPP	L2	99/N.D.	>20:1	96





[a] Reaction conditions unless otherwise specified: **1a** (0.10 mmol), **2** (0.15 mmol), catalyst (10 mol%), ligand (15 mol%) in THF (2.0 mL) at −20 °C for 24 hours. [b] Isolated yield. [c] Determined by ^{1}H NMR. [d] Determined by chiral HPLC. [e] $Cu(OAc)_2$ (5 mol%) and the other metal salt (5 mol%) were used. [f] The reaction was conducted at −50 °C. [g] ^{i}Pr$_2$O (2.0 mL) was used as solvent. [h] CuDPP refers to copper(I) diphenylphosphinate. [i] ^{i}Pr$_2$O (1.0 mL) was used as solvent.

inspection of different single metal salts indicated that the reaction catalyzed by $Cu(OAc)_2$ achieved exclusive γ-selectivity compared with other single metal salts (entry 4 vs entries 5–7). The different regioselectivities of these results illustrated that there might be stronger coordination between the copper and both the 1-alkynyl ketimine and α-substituted cyanoacetate compared with other metal salts (entries 1–7). Therefore, other reaction parameters were optimized using copper as the metal salt. Nucleophiles with ethyl, benzyl and *tert*-butyl ester groups were then tested (entries 8–10). Among them, a nucleophile bearing a bulky *tert*-butyl ester group was able to give the γ-addition product in 91% yield with higher stereoselectivity (entry 10). Gratifyingly, after optimization of the copper salts, solvent and temperature (for details, see Supporting Information), the asymmetric γ-addition reactions proceeded completely at −50 °C using copper(I) diphenylphosphinate (CuDPP) as the metal salt and ^{i}Pr$_2$O as the solvent, affording the target

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product in good enantioselectivity albeit with poor diastereoselectivity (entry 11).

Thereafter, ligands with different skeletons were investigated (for details, see Supporting Information). As a result, the benzyl substituted BOX ligand **L2** was proved to be the best ligand to promote the γ-selective addition reactions and afford the title product in 99 % yield with 95 % *ee* and >20:1 *dr* (entry 12). Finally, the reaction at a higher concentration afforded the chiral allene product in a slightly increased enantioselectivity with no influence on the reaction's reactivity and diastereoselectivity (entry 13). The absolute configuration of **3aa** was determined by X-ray diffraction (CCDC 2182161).

With the optimal reaction conditions in hand, the substrate scope of 1-alkynyl ketimines was explored first (Figure 2). The ketimines bearing methyl and *tert*-butyl groups at different positions of the phenyl ring were all amenable to the standard conditions, giving the corresponding products (**3ba**–**3ea**) in high yields (90–99 %) with excellent stereoselectivities (94–96 % *ee*, >20:1 *dr*). **3fa** with a *para*-substituted methoxy group was obtained in 99 % yield, 91 % *ee* and 18:1 *dr* at −30 °C. Next, the 1-alkynyl ketimines bearing electron-withdrawing groups at different positions of the phenyl groups gave the corresponding products (**3ga**–**3la**) in good to excellent yields (72–99 %) and enantioselectivities (94–98 % *ee*). However, it is obvious that the *dr* values of these allene products were reduced following a drop in the electron density of the phenyl group. This intriguing phenomenon could be attributed to the electron-withdrawing effect which might weaken the coordination between the 1-alkynyl ketimines and the copper catalysts, decreasing the diastereoselectivities. When R^1 is 1-naphthyl, 2-naphthyl or 2-thiophenyl, the expected products (**3ma**–**3oa**) were afforded in high yields (92–99 %) with good enantioselectivities (87–96 % *ee*) and diastereoselectivities (8:1–14:1 *dr*). To our delight, substrates with alkenyl substituents generated the corresponding products **3pa** in 92 % yield with excellent stereoselectivity (16:1 *dr*, 92 % *ee*). The ketimines bearing primary and secondary alkyl substituents were amenable to this catalytic system, affording the corresponding products (**3qa**–**3sa**) in medium to high yields (66–96 %) and with good stereoselectivities (90–95 % *ee*, 12:1–15:1 *dr*). Moreover, ferrocenyl substituted product **3ta** was gained in moderate yield (54 %) with good stereoselectivities (88 % *ee*, 10:1 *dr*) at −30 °C. Replacing ethyl ester to methyl ester had little influence on the reaction and

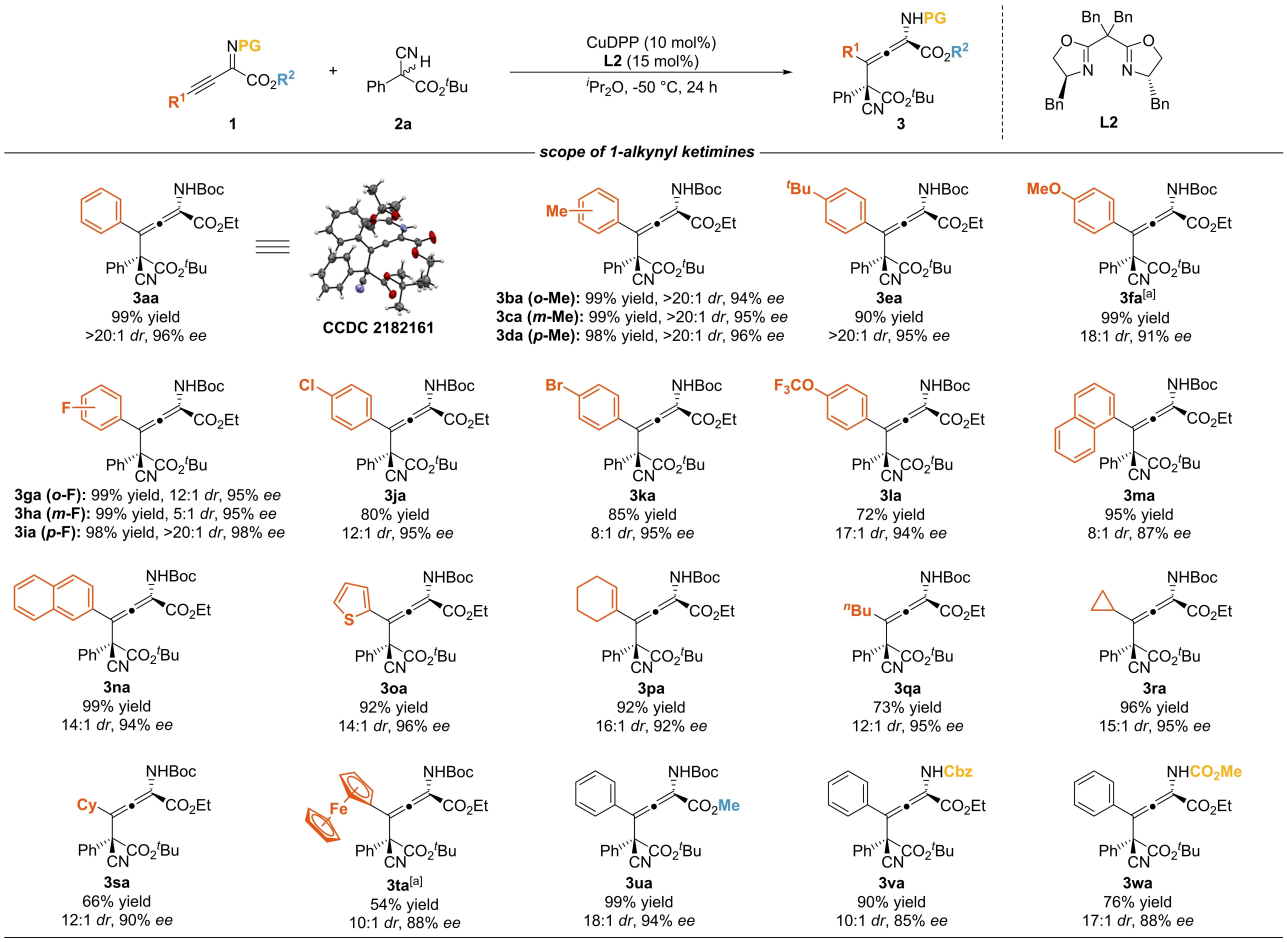


Figure 2. Scope of 1-alkynyl ketimines. Reaction conditions unless otherwise specified: **1** (0.10 mmol), **2a** (0.15 mmol), CuDPP (10 mol %), **L2** (15 mol %) in $^{i}Pr_2O$ (1.0 mL) at −50 °C for 24 hours; The yields refer to isolated yields; The *dr* values of **3** were determined by ^{1}H NMR; The *ee* values of **3** were determined by chiral HPLC. [a] The reaction was conducted at −30 °C.

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the desired product **3ua** was afforded in 99 % yield, 94 % *ee* and 18:1 *dr*. Finally, the 1-alkynyl ketimines protected by less bulky benzyloxycarbonyl (Cbz) and methyloxycarbonyl groups reacted with **2a** smoothly, providing products **3va** and **3wa** in good to high yields (76–90 %) with high diastereoselectivities (10:1–17:1 *dr*) and slightly reduced stereoselectivity (85–88 % *ee*).

As shown in Figure 3, the scope of nucleophiles **2** was then investigated. With different substituents at the *meta*- or *para*-position of the phenyl ring, the expected products (**3ab**–**3aj**) were all afforded in high yields (93–99 %) with excellent enantioselectivities (90–95 % *ee*) and diastereoselectivities (10:1–>20:1 *dr*). However, α-substituted cyanoacetates with either electron-donating groups or electron-withdrawing groups at the *ortho*-position provided extremely low reactivities under the standard conditions (for details, see Supporting Information). Only the *ortho*-fluorinated nucleophile can react with 1-alkynyl ketimines catalyzed by CuOAc at −30 °C and the desired product **3ak** was afforded in 51 % yield with 91 % *ee* and 10:1 *dr*. This result could be attributed to that the *ortho*-substitutents increased the dihedral angles between the phenyl ring of R^3 and cyano group, which improved the Gibbs energy of the deprotonated nucleophiles and enabled the nucleophiles difficult to approach ketimine substrates (For details, see Supporting Information). Furthermore, nucleophiles bearing naphthyl or heterocyclic groups gave the desired product (**3al**–**3an**) with good to excellent yields (91–99 %) and stereoselectivities (88–95 % *ee*, 15:1 *dr*->20:1 *dr*). Next, product **3ao** (R^4 = CO_2Et) was gained in excellent yield (99 %) yet with an obvious decrease in stereoselectivity (83 % *ee*, 3:1 *dr*), which indicated that the less bulky ester groups were unfavorable to the stereocontrol of this reaction. Finally, phosphoric ester modified product **3ap** was afforded in 77 % yield with enhanced control of stereoselectivity (99 % *ee*, >20/1 *dr*).

Considering the practicality of this methodology, the gram-scale synthesis of **3aa** was conducted (Scheme 2a). To our delight, the resulting product was afforded in quantitative yield with excellent stereoselectivity. The stereodivergent synthesis of molecules is meaningful since the absolute and relative configurations are influential for their biological activities in many cases. However, to the best of our knowledge, there have been no reports concerning the stereodivergent synthesis of tetrasubstituted allenes bearing a vicinal all-carbon quaternary center. Intriguingly, we found that an equivalent of MeMgBr could be employed as a base to deprotonate the carbamate NH group of **3aa** and facilitate the epimerization of **3aa**, which was an effective method for the stereodivergent synthesis of the product (for details, see Supporting Information). Thus, all four stereoisomers of **3aa** were obtained conveniently by the copper catalyzed asymmetric γ-addition reaction and the Grignard reagent promoted epimerization (Scheme 2b). The asym-

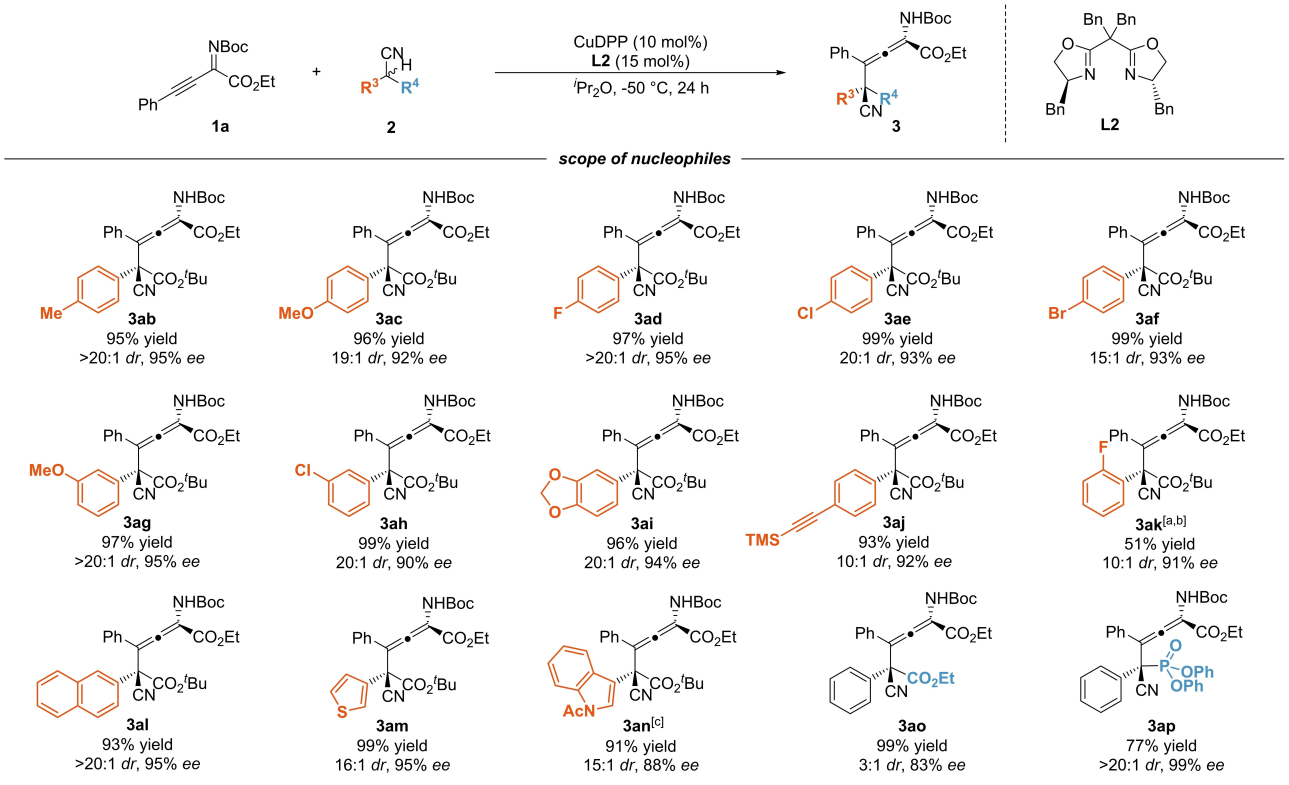
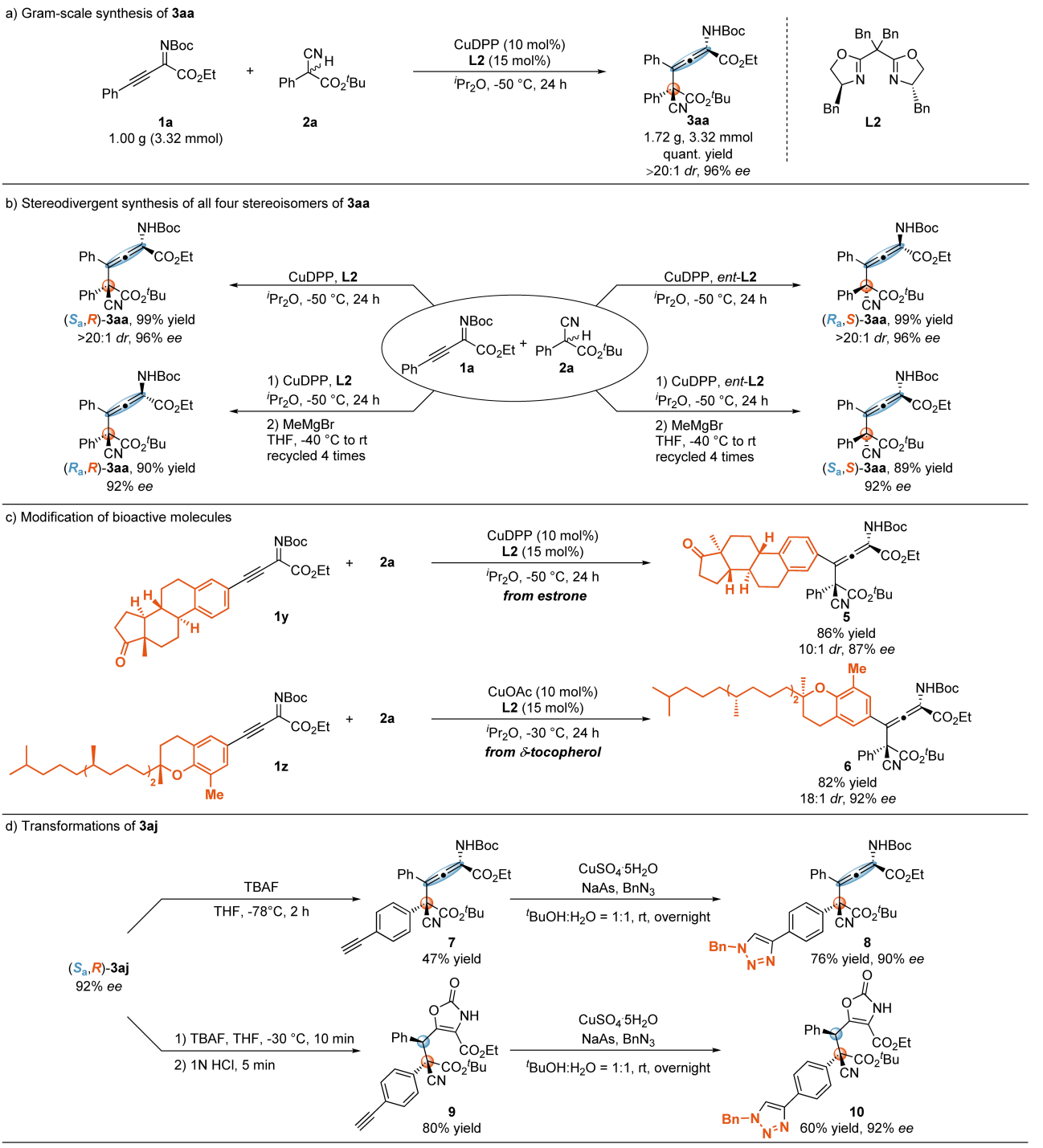


Figure 3. Scope of nucleophiles. Reaction conditions unless otherwise specified: **1a** (0.10 mmol), **2** (0.15 mmol), CuDPP (10 mol %), **L2** (15 mol %) in ^{i}Pr$_2$O (1.0 ml) at −50 °C for 24 hours; The yields refer to isolated yields; The *dr* values of **3** were determined by ^{1}H NMR; The *ee* values of **3** were determined by chiral HPLC. [a] The reaction was conducted at −30 °C. [b] CuOAc (10 mol %) was used as catalyst instead of CuDPP. [c] The reaction time was 48 hours.

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Scheme 2. Scale-up, stereodivergent synthesis and transformations of tetrasubstituted α-amino allenoates.

metric γ-addition reaction with the substrate **1y** derived from estrone proceeded smoothly under the standard conditions, giving the product **5** in high yield with good stereoselectivity (87 % *ee*, 10:1 *dr*) (Scheme 2c, top). Product **6** bearing a *δ*-tocopherol moiety was constructed successfully in 82 % yield with excellent stereoselectivity (92 % *ee*, 18:1 *dr*) (Scheme 2c, bottom). Additionally, using enantiomer of **L2** as the chiral ligand afforded the desired product with similar results, which indicates that the chirality of the bioactive moiety has little influence on the catalytic results (for details, see Supporting Information). The trimethylsilyl (TMS) group of **3aj** was readily removed by tetra-*n*-butylammonium fluoride (TBAF) at −78 °C, affording **7** in moderate yield. Then product **8** was synthesized in 76 % yield and with 90 % *ee* via a click reaction (Scheme 2d, top). Finally, the allene-based axial chirality of **3aj** was conveniently converted to central chirality forming an important type of oxazolone derivative **9** at −30 °C using

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TBAF, and the corresponding product **10** was obtained in 60 % yield and with 92 % *ee* by a subsequent click reaction (Scheme 2d, bottom).

To study the mechanism of this copper-catalyzed asymmetric γ-addition reaction, a series of control experiments were conducted (Figure 4). The asymmetric γ-addition using 2,2,6,6-tetramethyl-1-piperinedinyloxy (TEMPO) as a radical scavenger afforded the same result compared with the standard conditions, which manifested that the dual-copper-catalysis system might not be involved in a radical process (for details, see Supporting Information). Then, substrates **1aa** (R = Ph) and **1ab** (R = CF_3) were employed and no addition products were detected (Figure 4a). These results indicated the coordination between the ester group in **1a** and the copper catalyst plays an essential role in this reaction. In addition, during the solvent screening process, it was found that the reaction using acetonitrile as the solvent only afforded the α-addition product rather than the desired γ-addition product (for details, see Supporting Information). Thus, acetonitrile was used as a solvent instead of $^{i}Pr_2O$ at the standard conditions, and only a trace amount of γ-addition product was obtained with 64 % yield of α-addition byproduct (Figure 4b, condition A). Subsequently, 3.0 equivalents of acetonitrile were added to the reaction mixture under the standard conditions, affording the desired γ-addition product with an obviously reduced yield (75 %) accompanied by the α-addition byproduct (15 % yield) (Figure 4b, condition B). These may be due to the competitive coordination of the nitrile groups in acetonitrile and **2a** to the catalyst. Next, a reaction without **L2** was conducted at the standard conditions. As anticipated, almost no γ-addition product was obtained. and only the α-addition product **4aa** was afforded in 32 % yield (Figure 4b, condition C). These results suggested that **1a** and **2a** might coordinate simultaneously to one copper catalyst without ligands, in favor of the formation of the α-addition byproduct. Furthermore, non-linear effect experiments[15] and visual kinetic analysis[16] were conducted (Figure 4c and d). A positive non-linear effect between **3aa** and **L2** and an order of 1.8 with respect to Cu catalyst as observed by the visual kinetic analysis all suggested that this reaction was involved in a bimetallic process.

Subsequently, DFT calculations were utilized to explore the bimetallic catalysis (Figure 5). Firstly, the nucleophile **2a** coordinates with a Cu-BOX catalyst (**Cu*DPP**) to form intermediate **Int-1** via **TS1**. At the same time, the 1-alkynyl ketimine **1a** chelates with another **Cu*DPP** to form **Cu*-1a**. Then **Cu*-1a** attacks **Int-1** to afford an intermediate **Int-2**,

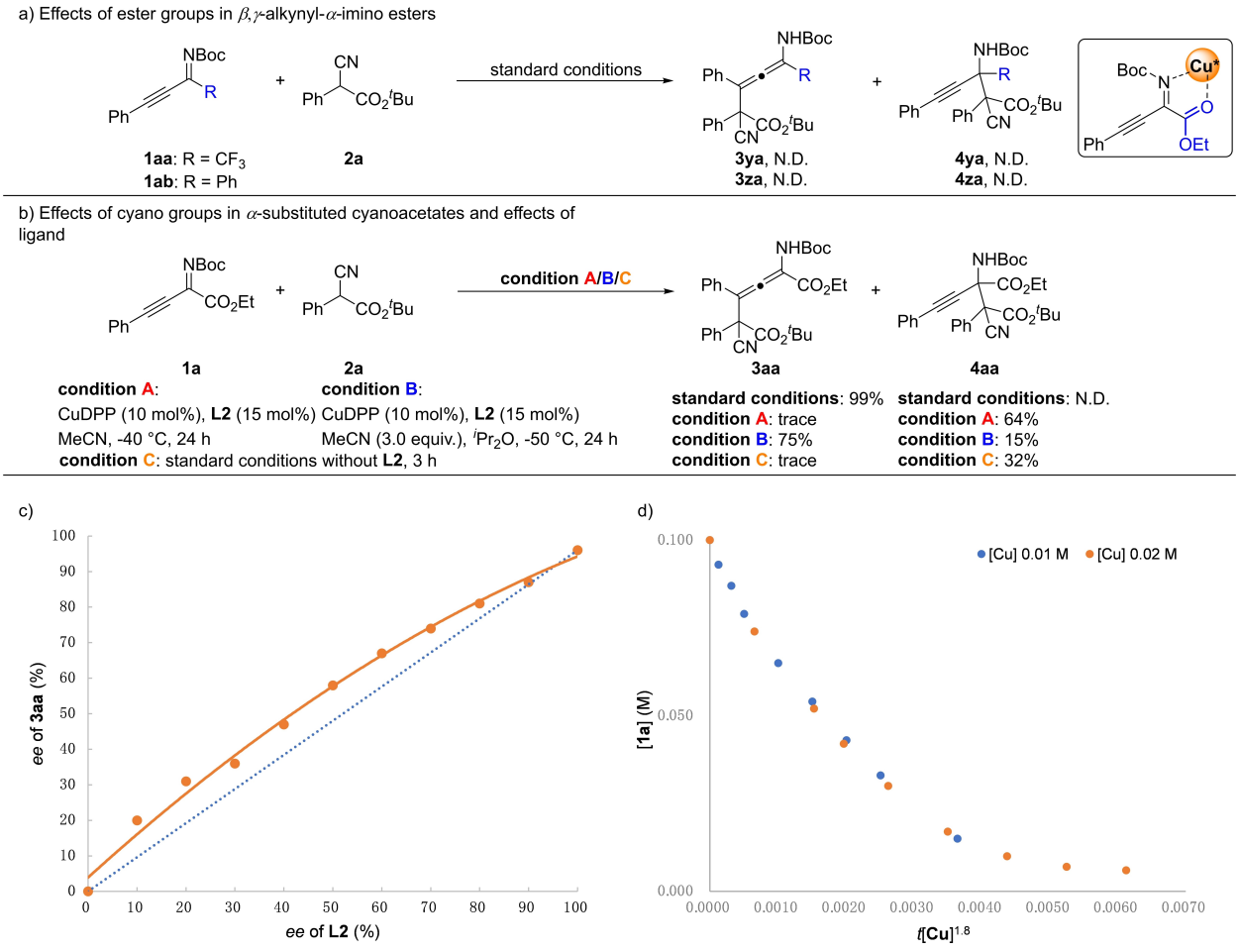
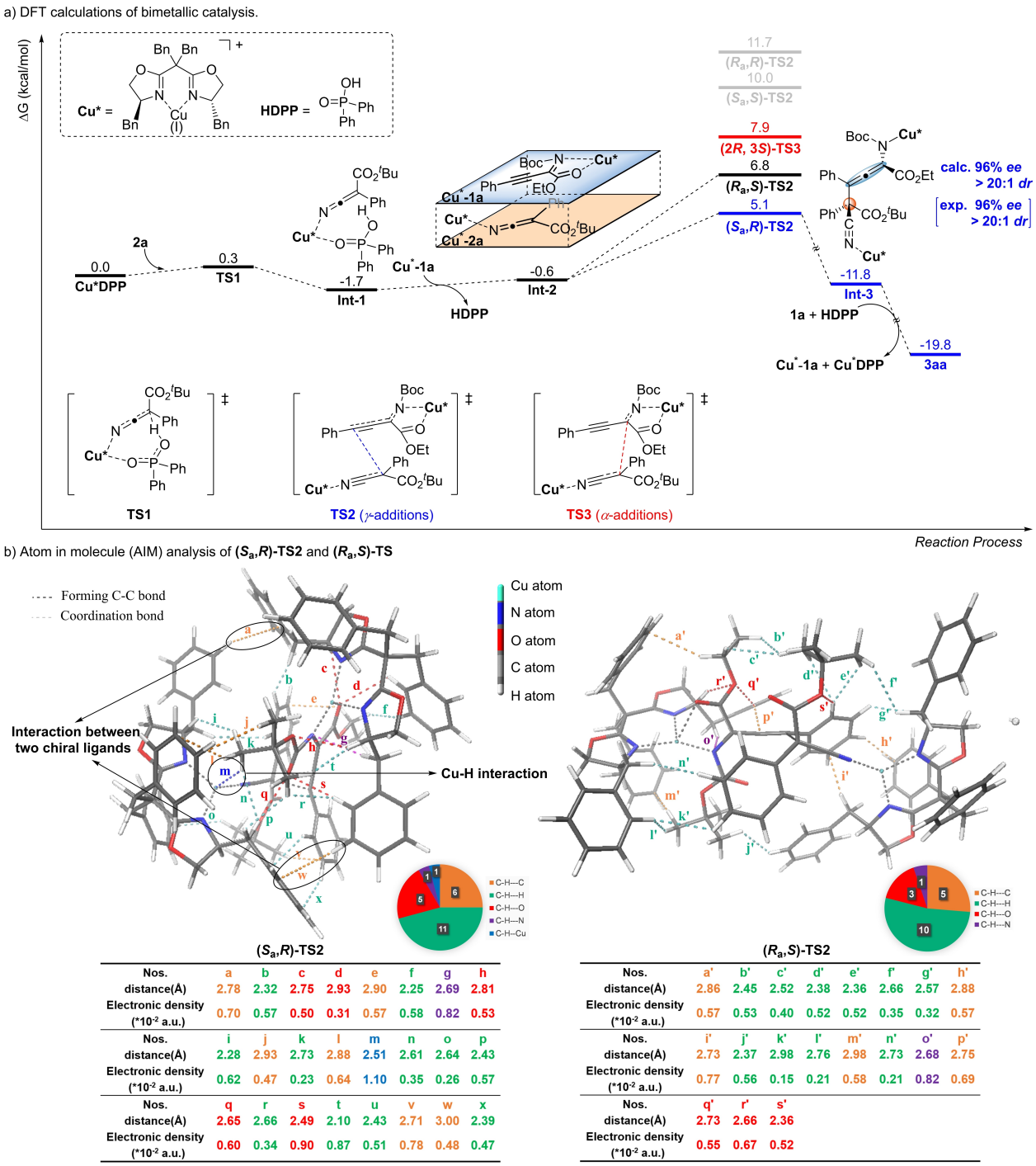


Figure 4. Mechanistic investigation.

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(Sa,R)-TS2								
Nos.	a	b	c	d	e	f	g	h
distance(Å)	2.78	2.32	2.75	2.93	2.90	2.25	2.69	2.81
Electronic density (*10^{-2} a.u.)	0.70	0.57	0.50	0.31	0.57	0.58	0.82	0.53
Nos.	i	j	k	l	m	n	o	p
distance(Å)	2.28	2.93	2.73	2.88	2.51	2.61	2.64	2.43
Electronic density (*10^{-2} a.u.)	0.62	0.47	0.23	0.64	1.10	0.35	0.26	0.57
Nos.	q	r	s	t	u	v	w	x
distance(Å)	2.65	2.66	2.49	2.10	2.43	2.71	3.00	2.39
Electronic density (*10^{-2} a.u.)	0.60	0.34	0.90	0.87	0.51	0.78	0.48	0.47

(Ra,S)-TS2								
Nos.	a'	b'	c'	d'	e'	f'	g'	h'
distance(Å)	2.86	2.45	2.52	2.38	2.36	2.66	2.57	2.88
Electronic density (*10^{-2} a.u.)	0.57	0.53	0.40	0.52	0.52	0.35	0.32	0.57
Nos.	i'	j'	k'	l'	m'	n'	o'	p'
distance(Å)	2.73	2.37	2.98	2.76	2.98	2.73	2.68	2.75
Electronic density (*10^{-2} a.u.)	0.77	0.56	0.15	0.21	0.58	0.21	0.82	0.69
Nos.	q'	r'	s'					
distance(Å)	2.73	2.66	2.36					
Electronic density (*10^{-2} a.u.)	0.55	0.67	0.52					

Figure 5. DFT calculations and atom in molecule (AIM) analysis of **(*S*$_a$,*R*)-TS2** and **(*R*$_a$,*S*)-TS2**.

releasing diphenylphosphinic acid (HDPP). **Int-2** exists as a stacked structure composed of **Cu*-1a** and **Cu*-2a** units. Subsequently, **Int-3** is formed via **TS2**. Then **Int-3** is protonated by HDPP and interacts with **1a** to produce the target product **3aa**, releasing **Cu*-1a** and **Cu*DPP** to join in the next catalytic cycle (Figure 5a). Among the four configurations, **(*S*$_a$,*R*)-TS2** has the lowest energy (5.1 kcal mol^{-1}). To our delight, the calculated *ee* and *dr* values of **3aa** (96 % *ee*, >20:1 *dr*) are consistent with the experimental results (96 % *ee*, >20:1 *dr*).

Compared with the γ-addition reactions, the asymmetric α-addition reactions catalyzed by bimetallic catalysts do not readily occur. The distortion energies of **Cu*-1a** and **Cu*-2a** were computed to explain the propensity of asymmetric γ-additions under bimetallic catalysis. Both **Cu*-1a** and **Cu*-2a** in **(*S*$_a$,*R*)-TS2** have much lower distortion energies compared with that of **(2*R*,3*S*)-TS3**, and the distance

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between the two Cu atoms in **(S_a,*R*)-TS2** (6.97 Å) is longer than that in **(2*R*,3*S*)-TS3** (6.50 Å), which demonstrates that **(S_a,*R*)-TS2** has a more expanded space compared with **(2*R*,3*S*)-TS3**, favoring the γ-addition reactions (for details, see Supporting Information). A possible process of mono copper catalyzed addition reactions was also investigated. It is interesting that the mono copper catalytic system prefers to afford α-addition products rather than γ-addition products (for details, see Supporting Information). Later, we studied the origin of the high enantioselectivity in the γ-addition reactions. The atom in molecule (AIM) analysis of **(S_a,*R*)-TS2** and **(R_a,*S*)-TS2** indicated that there are a great number of non-covalent intermolecular interactions in **(S_a,*R*)-TS2** (Figure 5b). Notably, the interactions between the two BOX ligands in **(S_a,*R*)-TS2** (**a** and **w**) contribute to the stabilization of **(S_a,*R*)-TS2**. The interaction of Cu–H (**m**) is significant for making Cu atom tetra-coordinated. Thus, the transition state **(S_a,*R*)-TS2** has much lower energy than **(R_a,*S*)-TS2**.

Conclusion

In conclusion, we have demonstrated an asymmetric γ-addition reaction of α-substituted cyanoacetates to 1-alkynyl ketimines with dual-copper-catalysis which was successfully applied to synthesize chiral tetrasubstituted α-amino allenoates bearing a vicinal all-carbon quaternary stereocenter in high yields and with excellent stereoselectivities (up to 99 % yield, 99 % *ee* and >20:1 *dr*). Mechanism studies revealed that the asymmetric γ-addition reactions are catalyzed by double chiral Cu catalysts. The asymmetric catalytic reaction was carried out smoothly on a gram scale and the stereodivergent synthesis of this type of compounds was firstly realized. In addition, the catalytic system was efficiently applied to introduce chiral tetrasubstituted allenes and all-carbon quaternary stereocenters into bioactive molecules. The allene-based axial chirality was transformed to central chirality and formed a significant type of oxazolone derivative.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Asymmetric Catalysis · Dual-Copper-Catalysis · Quaternary Stereocenter · Tetrasubstituted α-Amino Allenoate

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